

Network Analysis on Rumination

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2. Methods

2.1 Network analysis

Psychological networks have their own characteristics which differentiate them from other networks of the graph theory (S. Epskamp, Borsboom, et al., 2017b; Newman, 2010). Nodes usually represent some psychological measures or items of a psychological assessment tool, the edges of the network reflect the corresponding estimates of association derived from the data. Sample sizes have direct impact on estimated accuracy and stability of the psychological network, thus the accuracy of the network parameters and measures is crucial: By obtaining robust results we can have a more realistic picture of a network without knowing the true network shape and structure. Just like any other psychological variables, network characteristics are also sensitive to sampling, estimation methods, included variables, outliers and various other components of an analysis. As (S. Epskamp, Borsboom, et al., 2017a) points out, psychological networks differ from *reflective latent variable models* by not seeking for a common cause of co-occurrence of variables (such as symptoms or different items of a questionnaire), but instead their focus is on quantifying and visually show the multivariate dependencies and the different pathways of direct relationships between the nodes. A major conceptual difference emerges from the network perspective for psychological data: psychological variables, such as symptoms, states, attitudes and other constructs, (represented as nodes) have direct impact on each other, and are not caused by some latent variables (S. Epskamp & Fried, 2018). In the current study, we used a fairly large sample as shown above. Before going further into network modeling, we estimated the 95% confidence intervals of network centrality measures to have an understanding of the sampling distribution of our estimates. According to S. Epskamp, Borsboom, et al. (2017a) low sample size and non-regularized estimates might lead to amplify any bootstrap bias; by having a larger sample of individuals, we expect lesser levels of bias of any bootstrap estimates for network accuracy and stability.

2.1.1 Removing redundant items As a preparation step, we applied the *goldbricker* function from the R package *networktools* (Jones, 2017) to search for pairs of items in the RRS, SCRS and their combined networks that were highly correlated ($r > 0.50$) and showed highly the same pattern with other items. When such redundant pairs were identified, reduction techniques were put into use (with the *net_reduce* function in R): instead of the pair, the first principal component (PCA) of the two variables were kept in the data as a new, merged variable.

The RRS items required no reduction steps as the highly correlating items did not extend their pattern to more nodes (their correlations to other nodes did not violate the criteria that $> 75\%$ of these correlations did not differ significantly).

2.1.2 Network Analysis with LASSO In order to map the relationships of the actual RRS and SCRS items, both on their respective networks, and in a combined network, we used the LASSO algorithm (Friedman et al., 2007). The LASSO uses L1 regularization, and results in a partial correlation network where

edges represent direct associations between pairs of items, while the influence of all other items in the network are kept controlled. The regularization by L1 penalty ensures that trivially small partial correlations are set to zero (McNally et al., 2017), thus ensure a *sparse* network (S. Epskamp & Fried (2018)). The tuning parameter of *gamma* in the graphical LASSO algorithm has been set to 0.5 value in accordance with (McNally et al., 2017) and (Bernstein et al., 2019) in order to achieve sparse graphs. The final networks were chosen by the lowest values of the *extended Bayesian Information Criterion* (EBIC).

2.1.3. Community detection In order to get an understanding of community of nodes and to see if nodes can be grouped into neighbourhoods, we used the *igraph* R package. When detecting possible communities of nodes, the detection happened with the following parameters in the *spin glass algorithm* (Bernstein et al., 2019; Reichardt & Bornholdt, 2006).

2.1.4. Expected Influence of nodes To understand certain items' centrality and importance, besides regularly used centrality measures (such as strength, closeness and betweenness of nodes in a network), one can also use the One- and Two-Step Expected Influence (EI) measures (Donald J. Robinaugh, 2016). Traditionally in unweighted networks (where network edges are encoded to 0 or 1) a node's centrality is the sum of edges, while in weighted nodes it is the sum of the edge weights. In this approach negative edges are treated the same way as positives, as the absolute values are used for centrality estimation. On the other hand, the Expected Influence metrics take into account if a node has positive and negative edges, too (Heeren et al., 2018).

2.1.5. Identifying important nodes In order to further elaborate the question of node importance within the network, we identified *bridge nodes* using the functions provided by the *networktools* R package (Jones, 2017). We estimated both one-step and two-step bridge expected influence measures which reflect the sum of edge weights connecting one node to other groups on nodes (or communities of nodes); and also express a node's secondary influence on the rest of the network, respectively. Higher influence scores indicate that a certain node is more likely to active neighbouring nodes, subgroups of nodes or communities.

2.1.6. Network accuracy and stability Accuracy and stability have an important role in estimating network parameters when samples might vary in characteristics and size (S. Epskamp, Borsboom, et al., 2017a, 2017b). One way of estimating network accuracy is to produce a 95% confidence interval of edge-weights with some variant of bootstrap. In our case we chose the non-parametric variant because of interpretability. Network stability is quantified with the *Correlation Stability* (CS-coefficient): an estimate of the maximum proportion of cases that can be dropped (with 95% certainty) to retain a correlation with the original centrality of higher than 0.7. Simulation studies (S. Epskamp, Borsboom, et al., 2017b) show that the CS-coefficient should be between 0.25 and 0.5.

3. Results

3.1 Network plots

When estimating networks, we modeled RRS, SCRS and their combined network as Gaussian graph-networks which are networks of partial correlation coefficients (S. Epskamp, Waldorp, et al., 2018). In many parts of this analysis we also utilized functions from the *qgraph* R package (Epskamp, 2012). Please refer Fig. 1-3.

3.2 Community detection

Community mapping of RSS and SCRS with the *spin glass algorithm* (Bernstein et al., 2019; Reichardt & Bornholdt, 2006).

RRS items (Fig.4.) are grouped into two communities. In one we find the strongly connected items *alone1* and *alone2* with items *write*, *person* and *analyze*. The rest of the items constitute the other main community.

The self-critical rumination items exhibit one large community as a whole (Fig.5.).

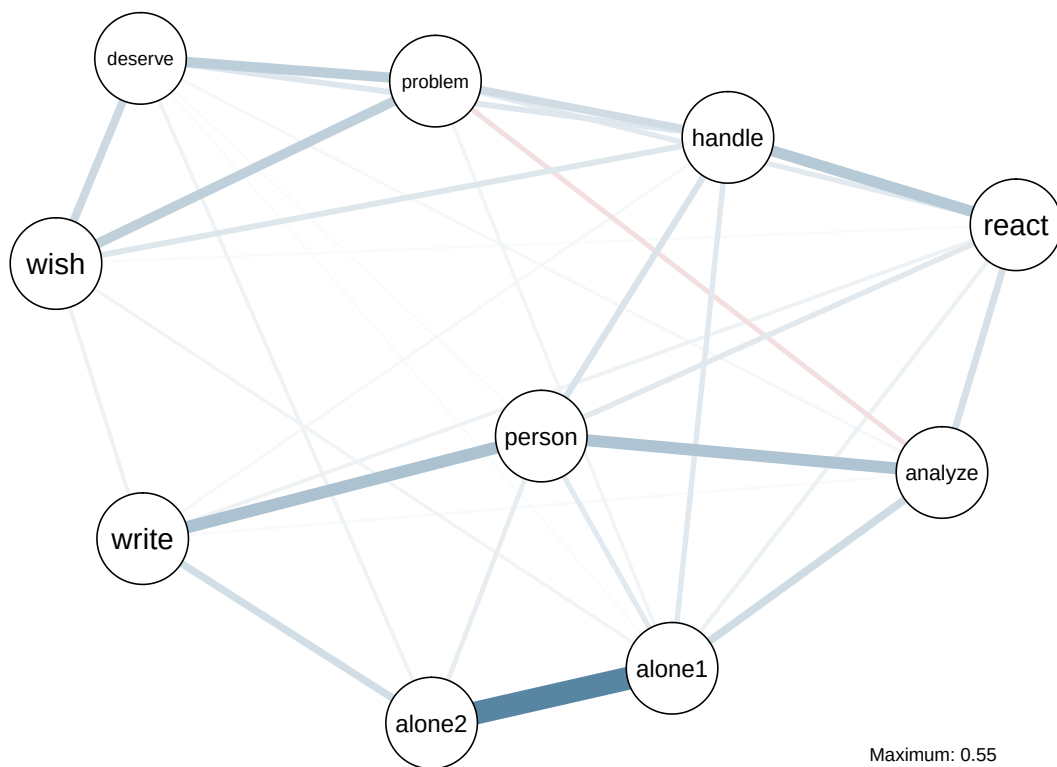


Figure 1: RRS network with gLASSO

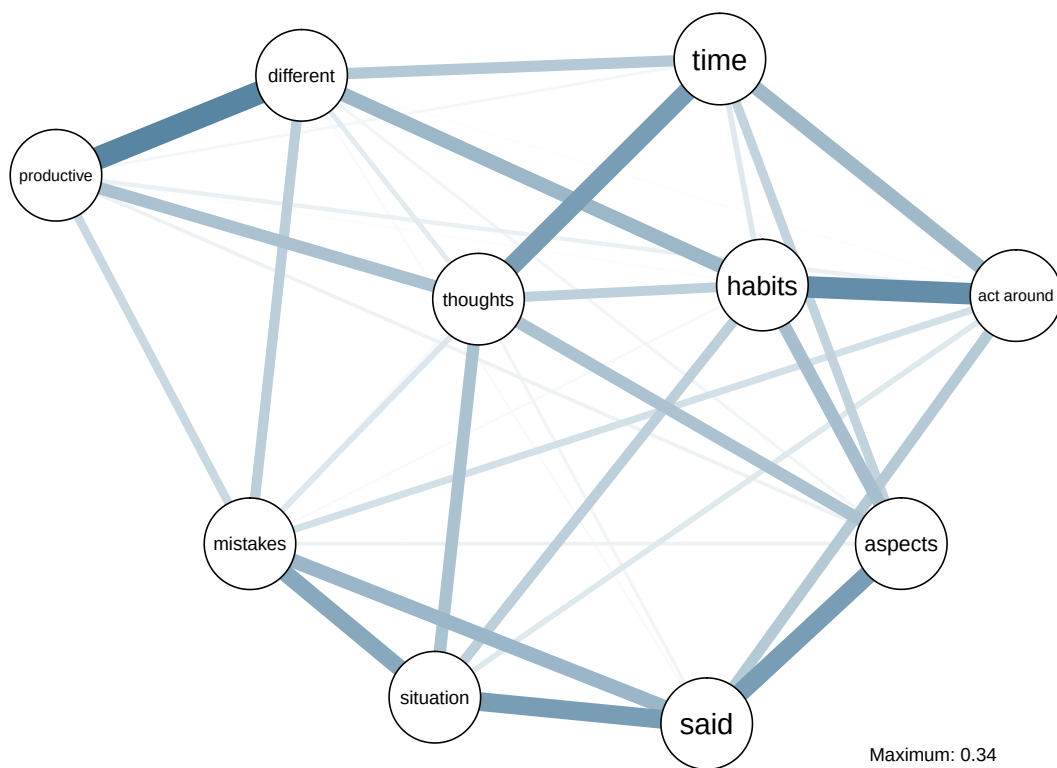


Figure 2: SCRS network with gLASSO

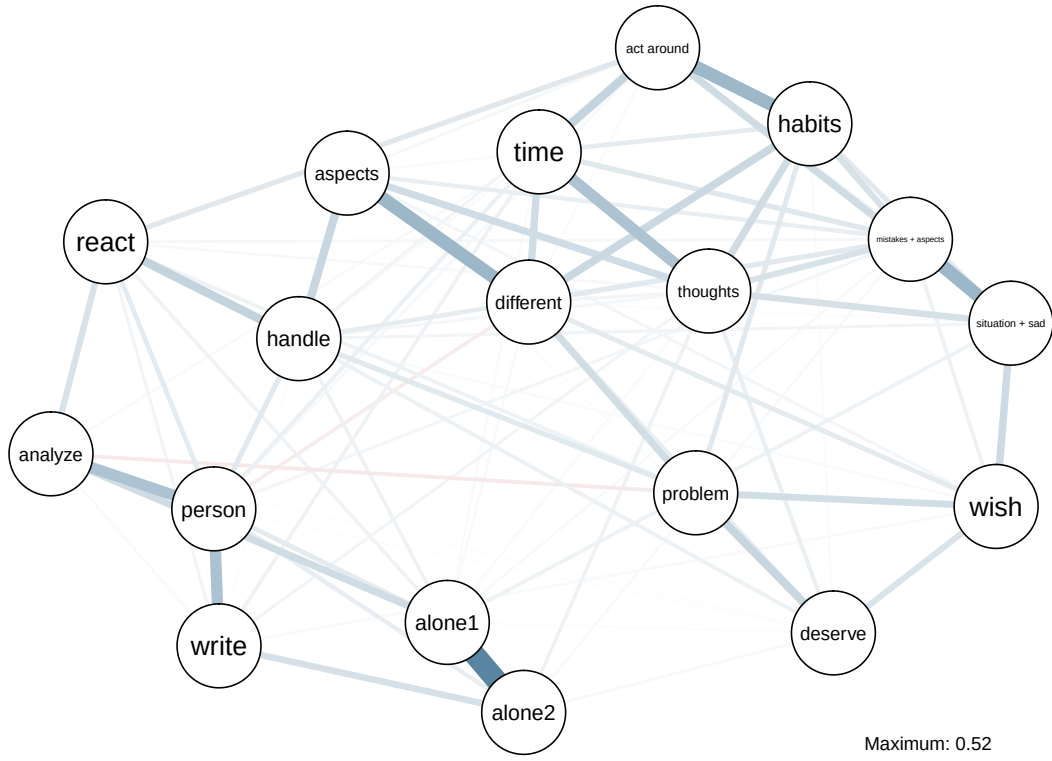


Figure 3: Combined RRS and SCRS network

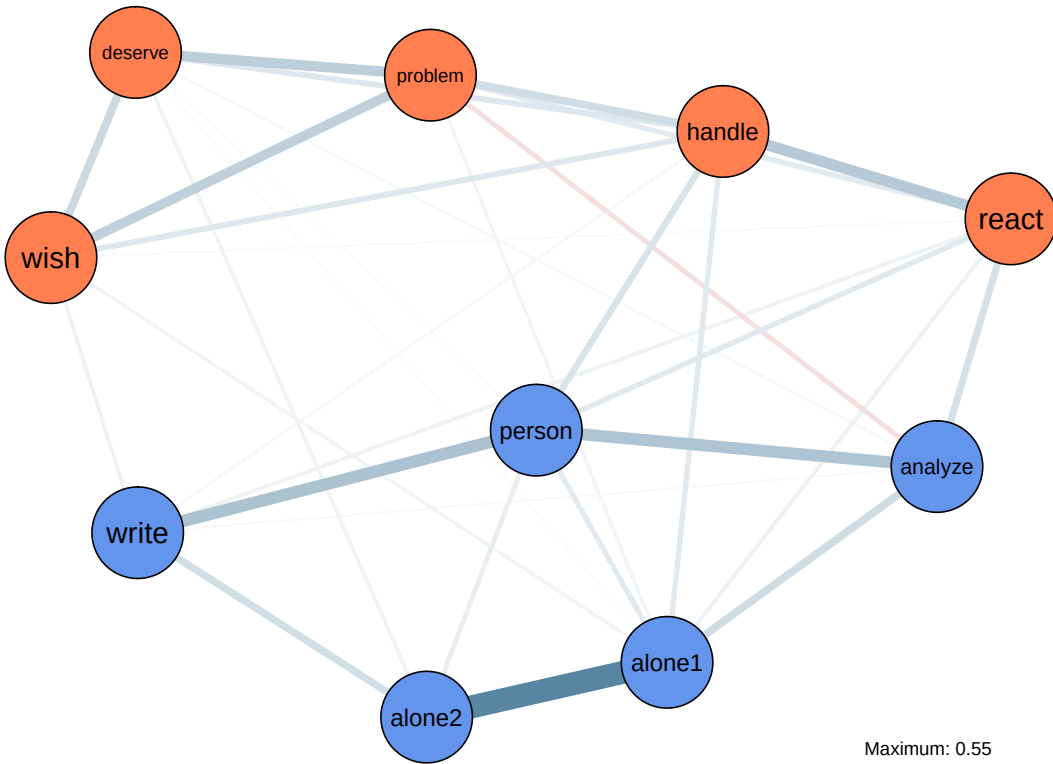


Figure 4: RSS communities identified with parameters $\gamma = 0.5$, $\text{spins} = 25$, $\text{starting temperature} = 1$, $\text{ending temperature} = 0.01$, $\text{cooling factor} = 0.99$

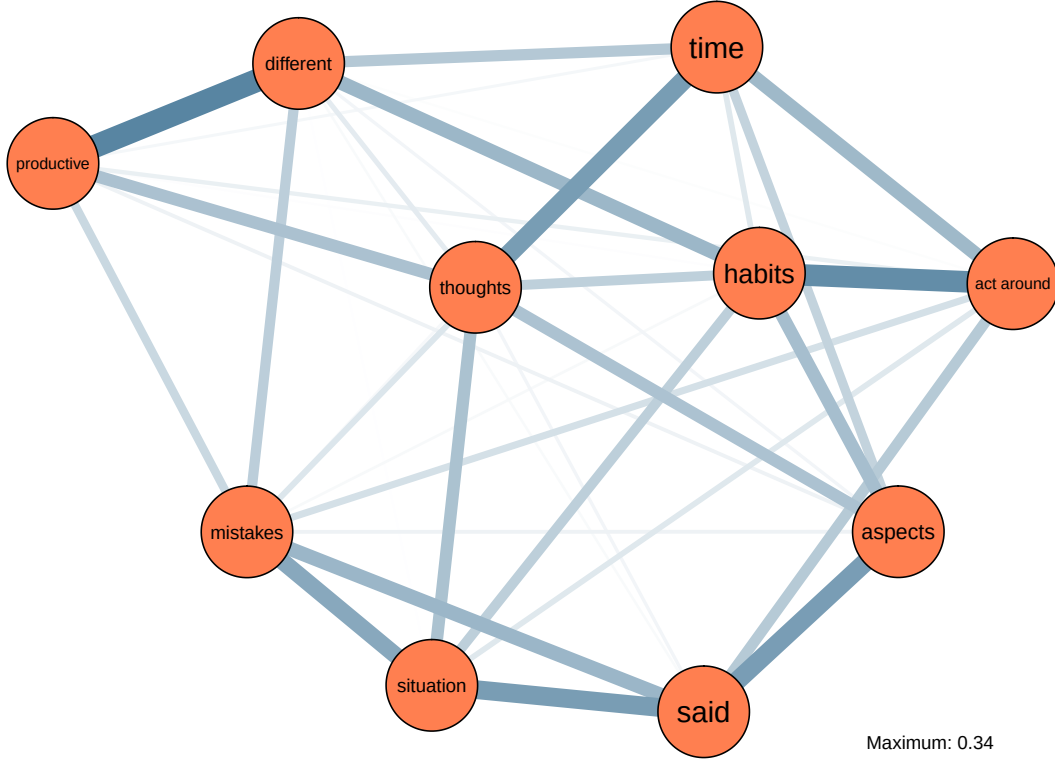


Figure 5: SCRS communities identified with parameters $\gamma = 0.5$, $\text{spins} = 25$, starting temperature = 1, ending temperature = 0.01, cooling factor = 0.99.

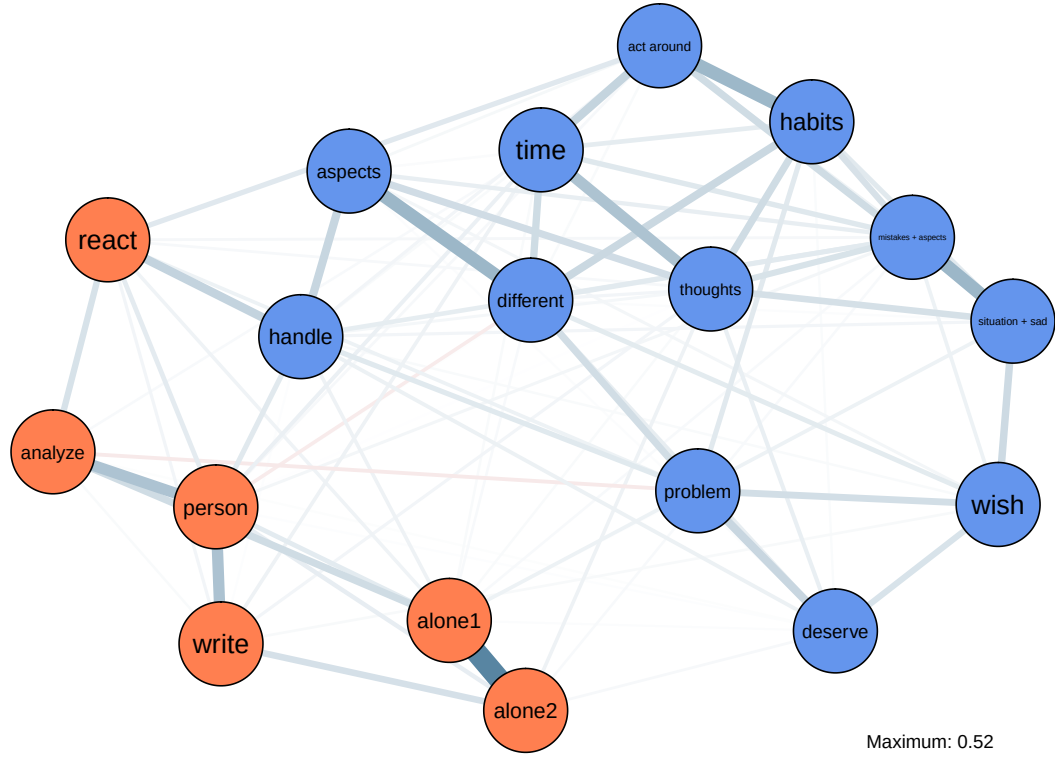


Figure 6: RRS and SCRS combined communities identified with parameters $\gamma = -1.5$, $\text{spins} = 25$, starting temperature = 1, ending temperature = 0.01, cooling factor = 0.99.

When all RRS and SCRS items are handled as one covariance and one Gaussian graph-network, 4 items of RRS (*handle*, *problem*, *deserve* and *wish*) are grouped into the community of the SCRS items (Fig.6.).

3.3 Expected Influence

In order to estimate the Expected Influence metrics, we used the *expectedInf* function of the *networktools* R package. The influence score are standardized.

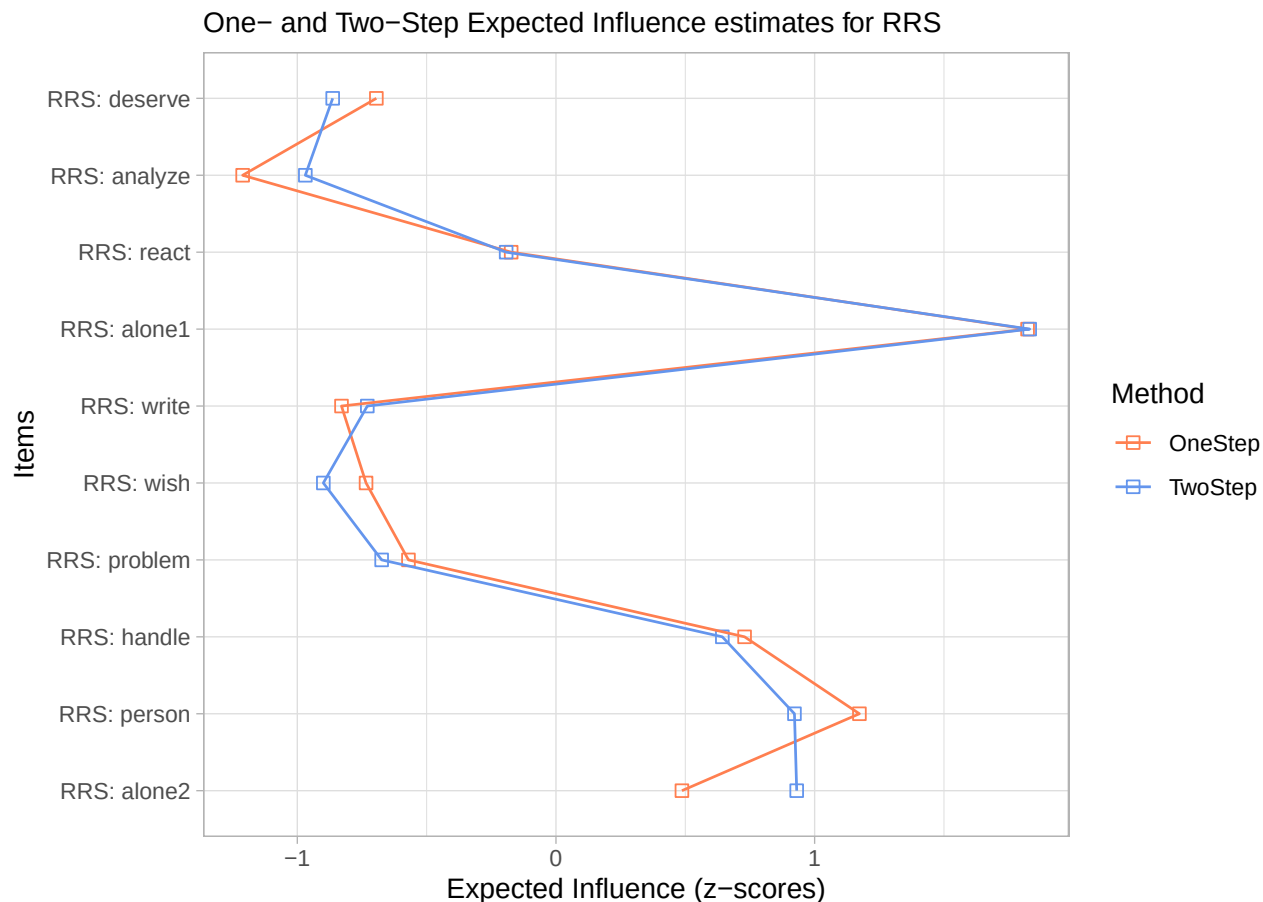


Figure 7: RRS Expected Influence Measures

Regarding RRS influence estimates (Fig.7.): Items such as *alone1*, *alone2* and *person* are high in terms of expected influence, meaning that these items play more central role in the RRS network, by changing these nodes, the rest of the network changes more likely.

SCRS-items (Fig.8.) such as *habits* and *thoughts* are more influencing nodes than *productive*.

In the combined network scenario (Fig.9.), the RRS item *alone1*, the SCRS items *habits* and *thoughts* maintain their influencing position among the items. Due to the combination of tools, the SCRS item *different* has increased its influence, and the items *mistakes* and *aspects* have increased their impact as a PCA component. This also means that in the combined network they proved to be too redundant and the *goldbricker* algorithm used their PCA-component instead.

3.4 Bridge nodes

Using the *networktools* R library bridge expected influence measure were estimated for the three tools we used in this study. We also added aesthetics to reflect which items belonged to the same community according to our previous clustering steps (Fig. 10-12.).

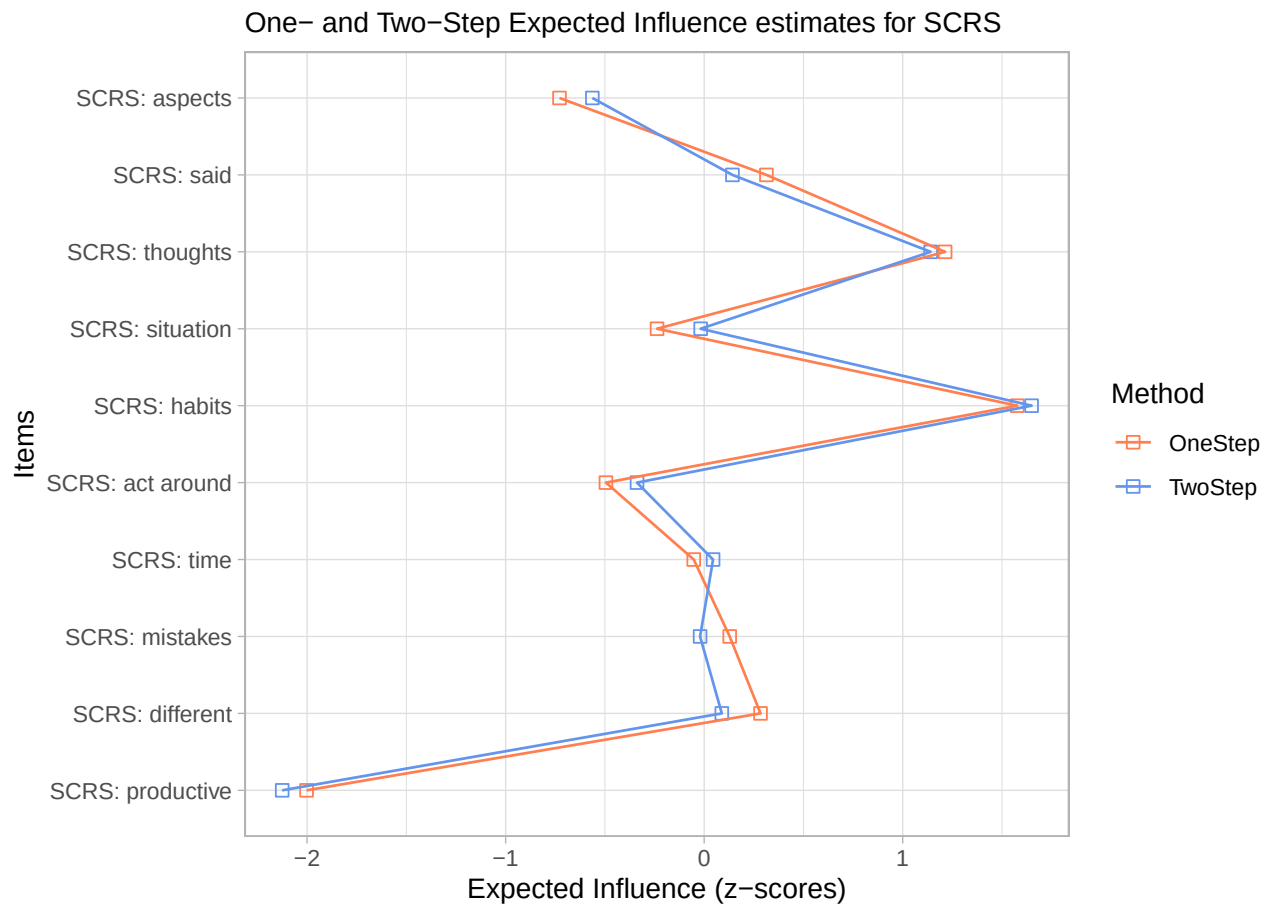


Figure 8: SCRS Expected Influence Measures

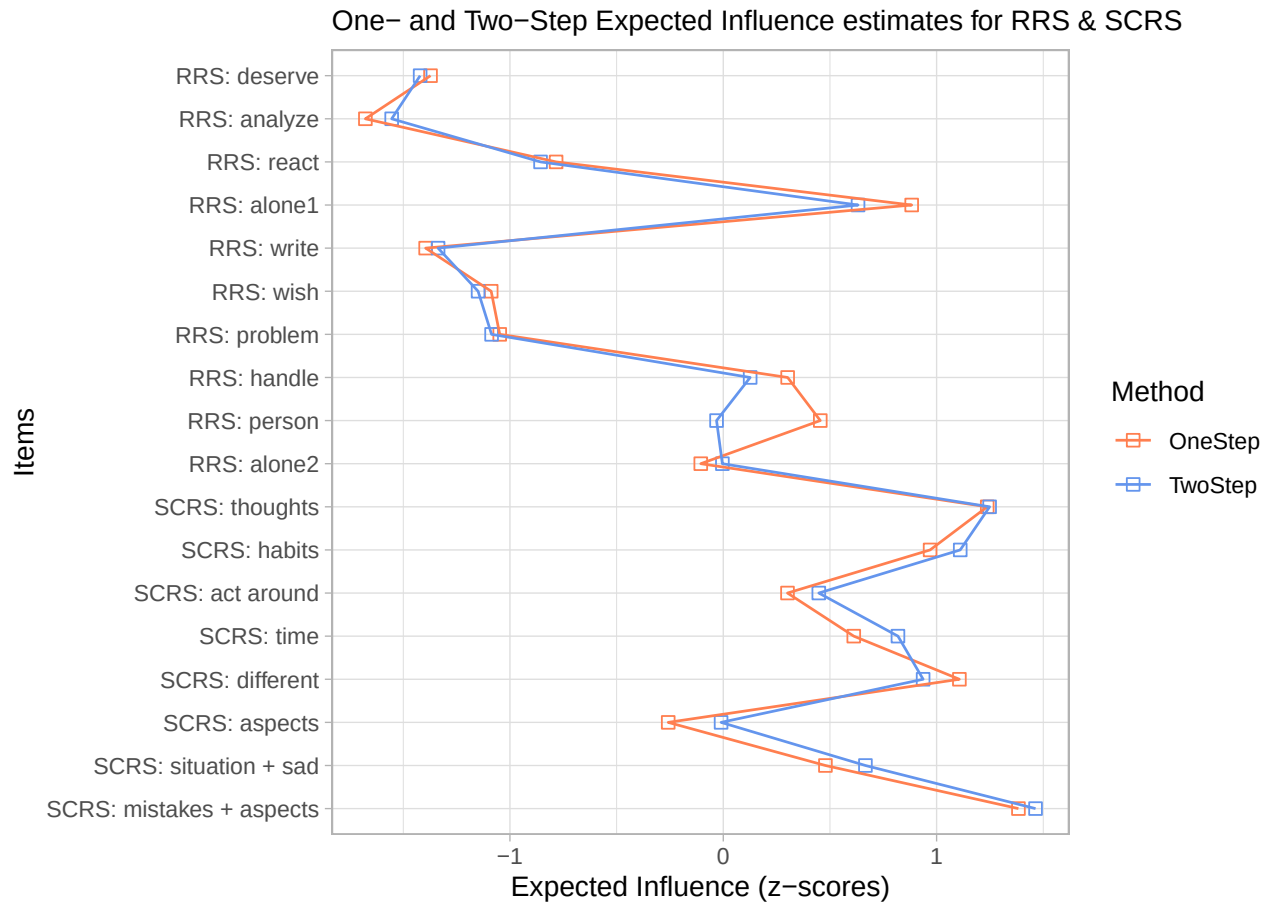


Figure 9: Combined RRS and SCRS Expected Influence Measures

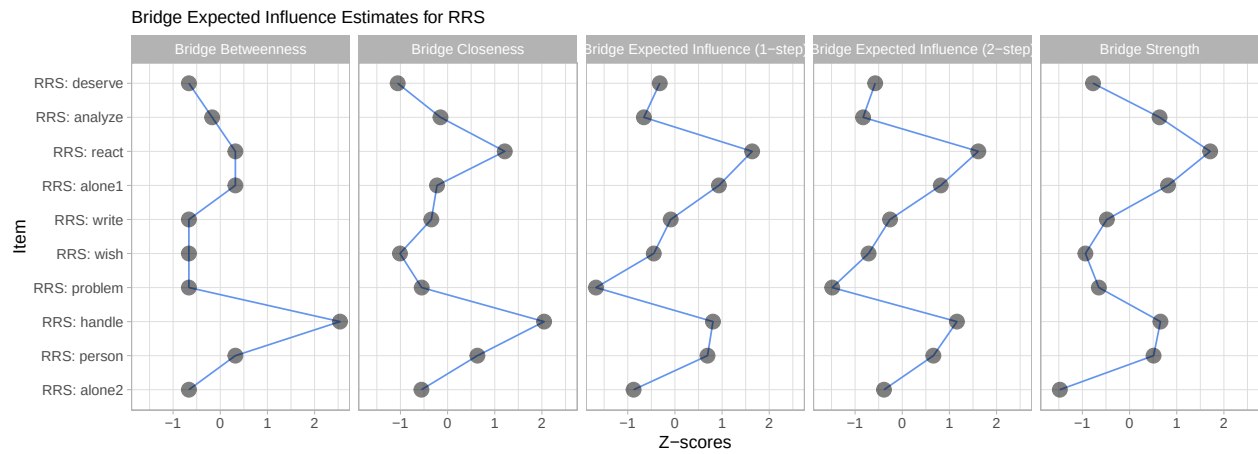


Figure 10: RRS bridge estimates

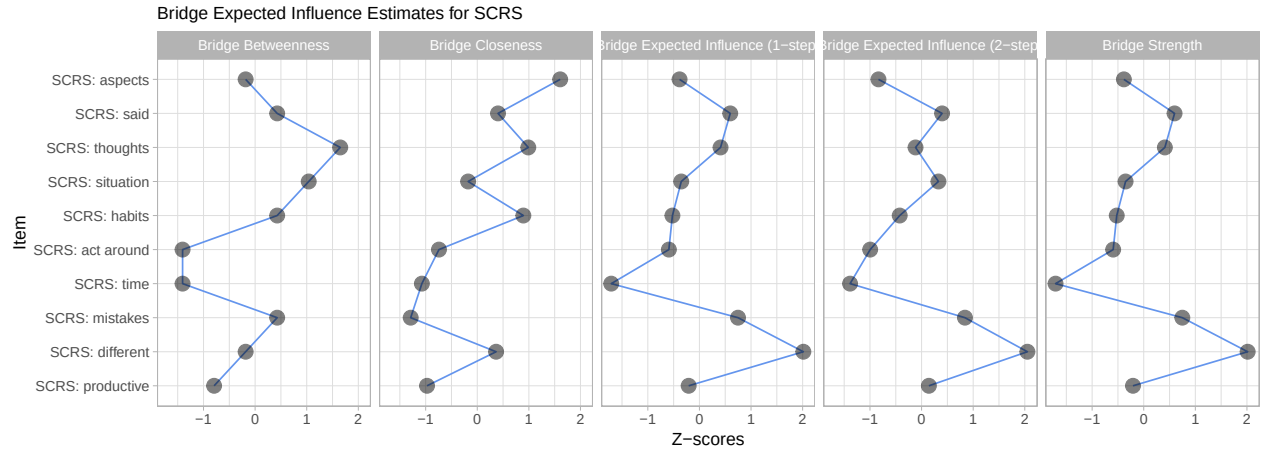


Figure 11: SCRS bridge estimates

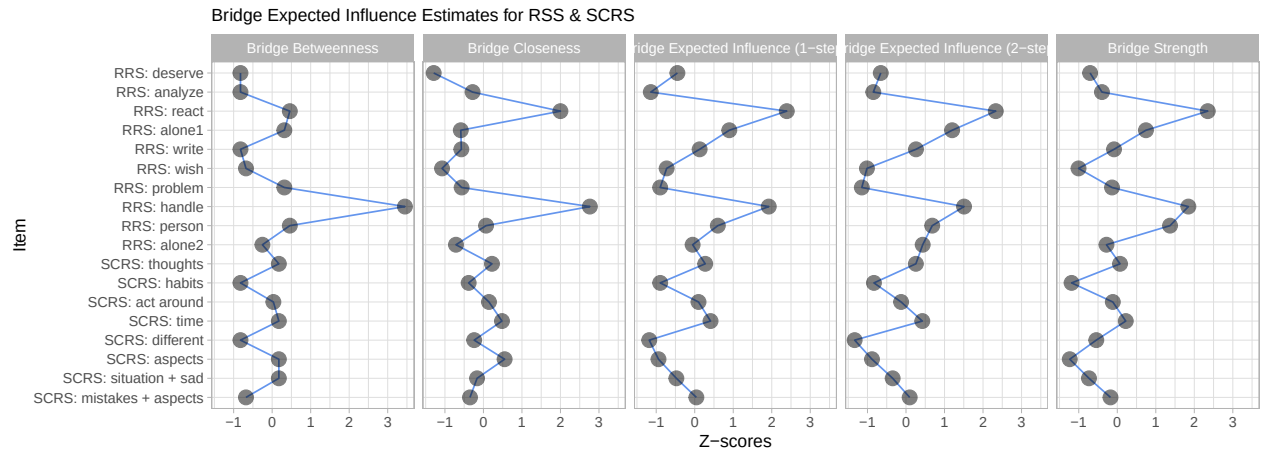
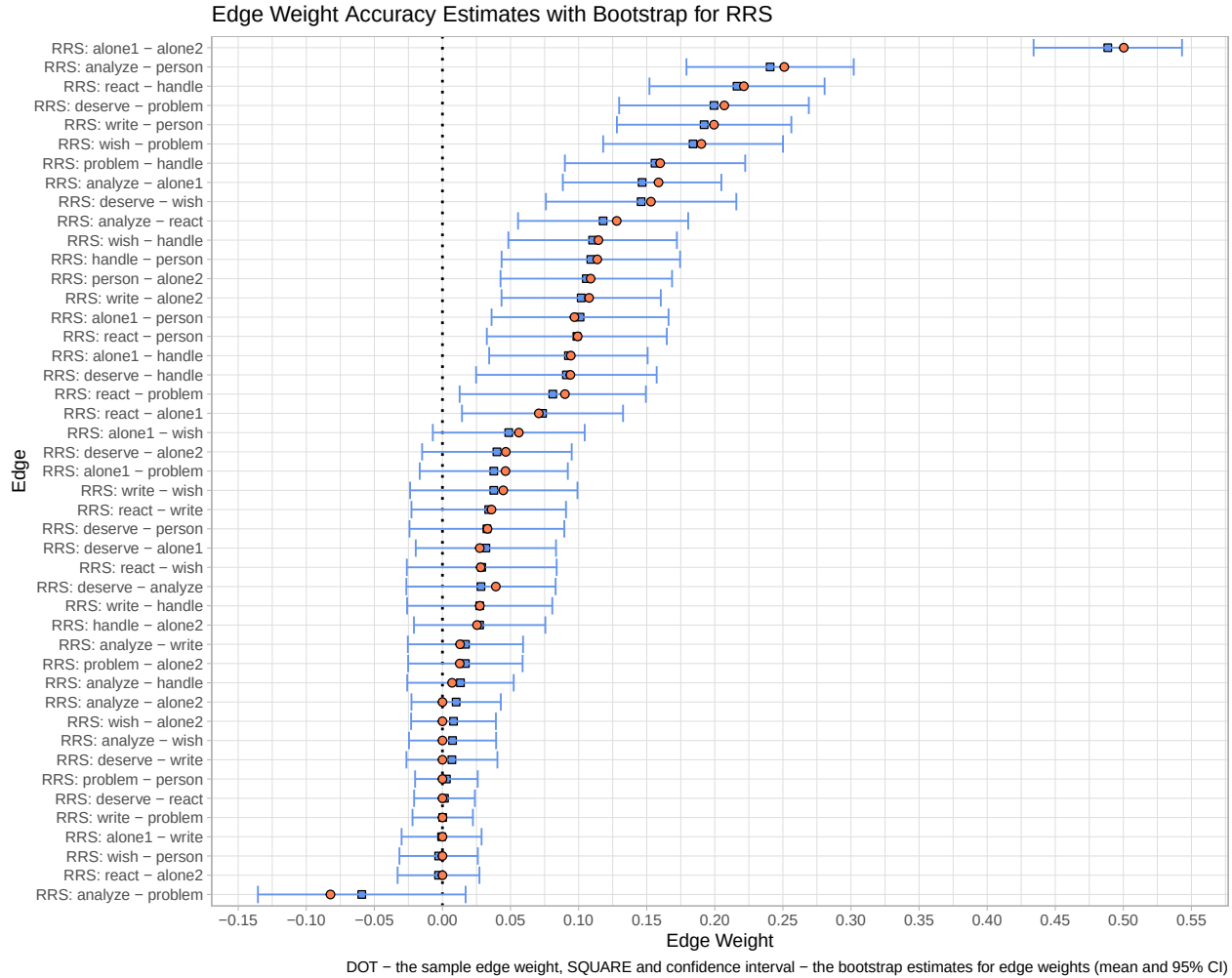


Figure 12: Combined RRS and SCRS bridge estimates

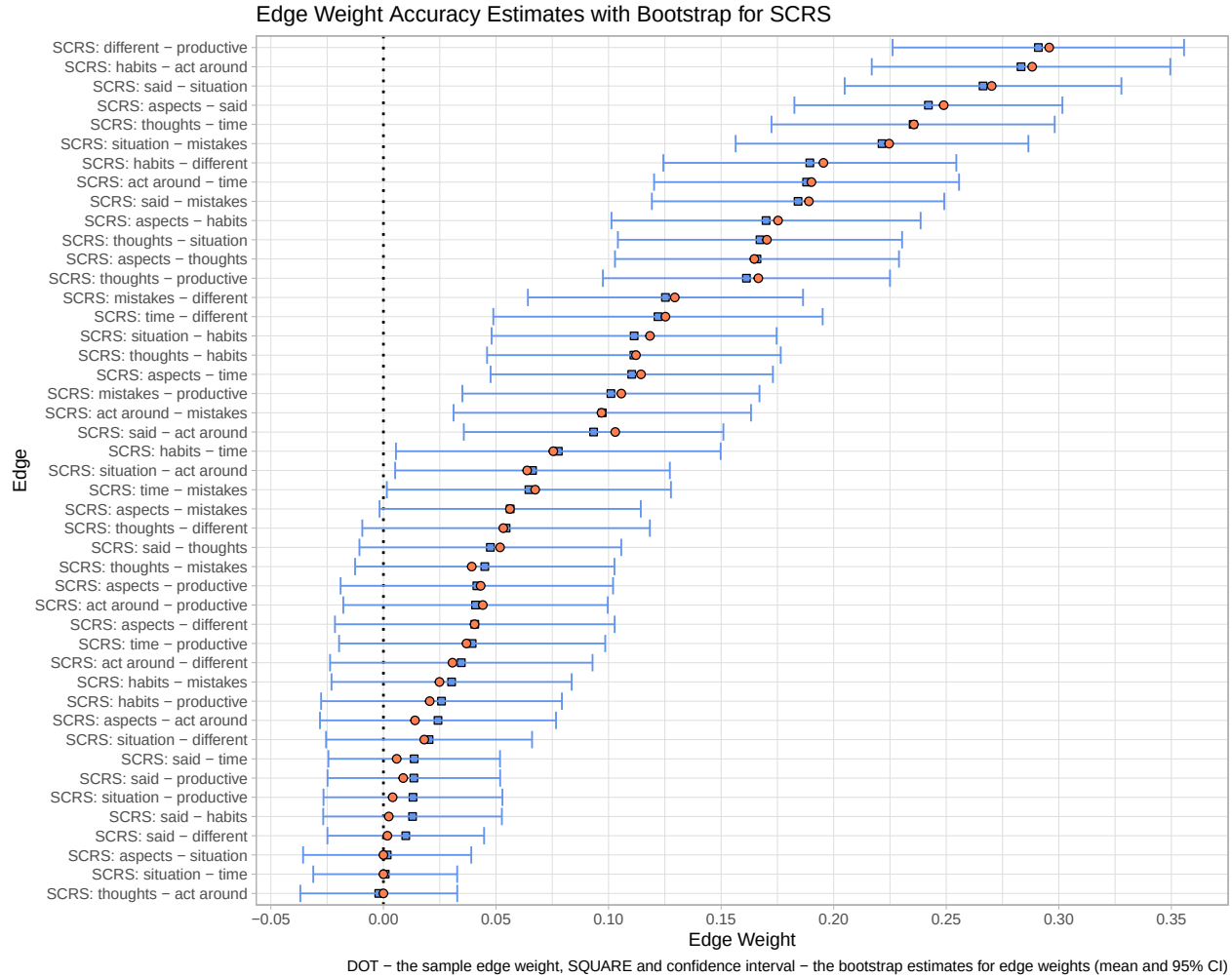
3.5 Network accuracy and stability

We used the *bootnet* R package to have an estimate of edge-weight accuracy with *non-parametric bootstrap* (2500 times sampling) and also produce measures of centrality stability with *case dropping bootstrap* (to simulate cases where only the subset of the existing data would be available).

Network accuracy plots show pairs of nodes (pairs of items) by ascending order of the edge-weights. 95% confidence intervals and the vertical line of 0 edge-weight are shown separately.

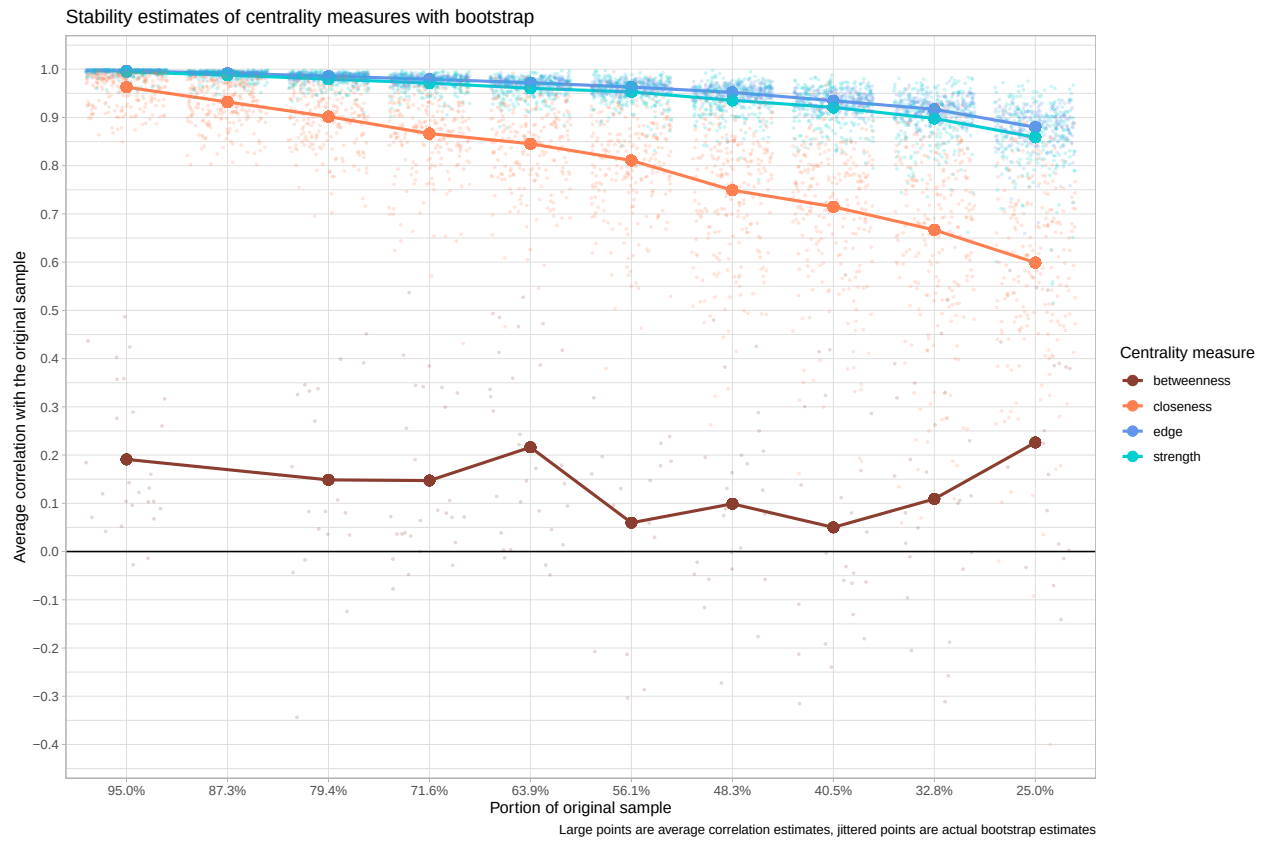


From the pair of items *react* and *alone1* confidence intervals of edge weights start to incorporate the value of 0 and thus we cannot be sure if their weight persist given different samples and varying sample characteristics.

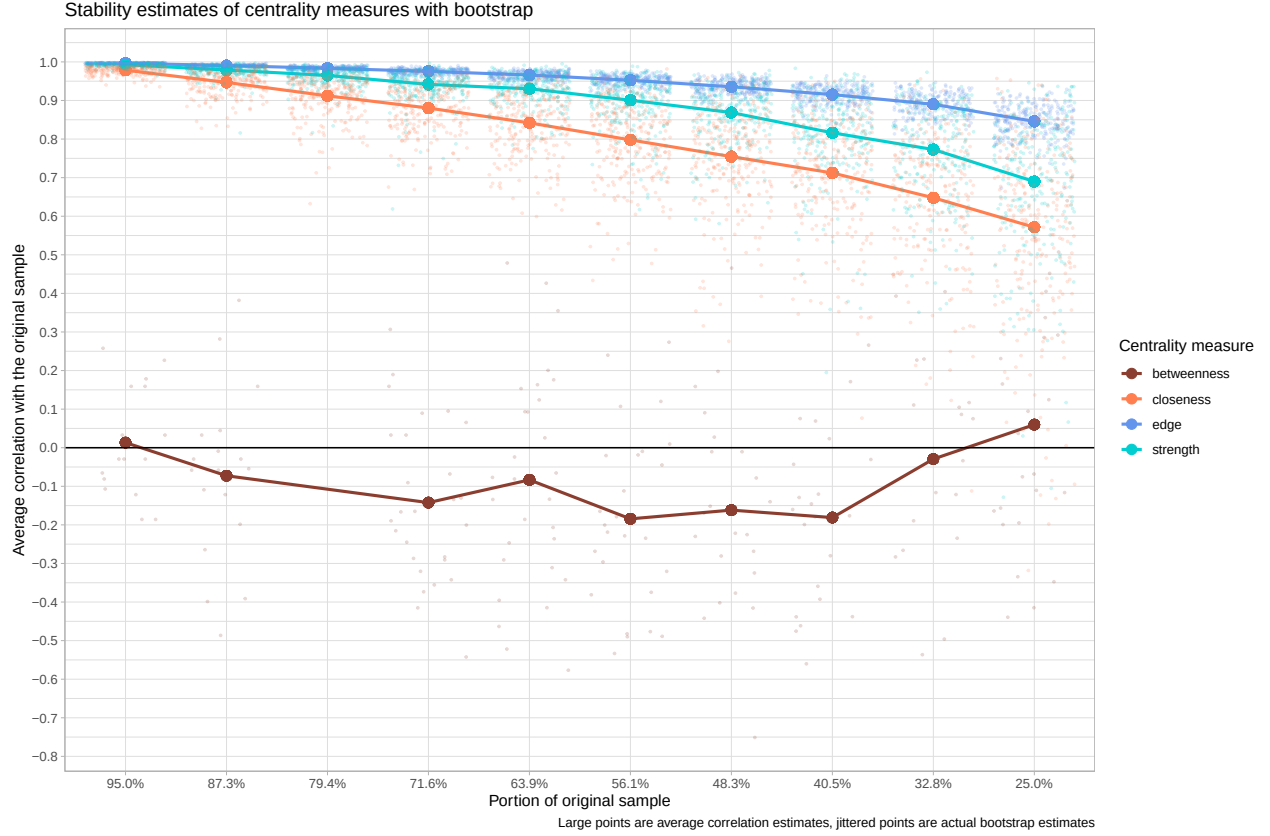


Similar to RRS, SCRS item pairs are ordered by their edge-weight estimates. Below the pair of items of *time* and *mistake*, accuracy might not be non-zero under different sampling conditions. The combined RRS and SCRS network estimates are added to the *Appendix* section of this paper.

Network stability estimates for our three networks were carried out by using the *bootnet* function from the R package bearing the same name. We performed a *case dropping bootstrap* to simulate scenarios where fewer and fewer cases of the original sample would have been available and quantified the stability of the centrality metrics with the *Correlation Stability* (CS) coefficient. When estimating bootstrap, we set the number of bootstraps to 2500.



Edge weights and node strength estimates exhibit strong stability, whereas closeness estimates are more sensitive to sample size drops. Betweenness scores seem to be insensitive to sample sizes and are consistently weak in terms of their low correlations with the original estimates for RRS.



Edge weights, node strength and closeness estimates exhibit strong stability, while betweenness scores seem to be insensitive to sample sizes and are consistently weak in terms of their low correlations with the original estimates for SCRS, similar to RRS.

Table 1: Correlation Stability coefficient estimates for RRS, SCRS and their combined network

Metric	RRS	SCRS	combined
betweenness	0.0495	0.1273	0.0000
closeness	0.2838	0.2838	0.5169
edge	0.7500	0.7500	0.7500
strength	0.7500	0.5169	0.7500

As shown at (S. Epskamp, Borsboom, et al., 2017b), the CS-coefficient is ideally between 0.25 and 0.5, the latter is the cut-off point for the estimate. The results show that RRS and SCRS closeness metrics are stable in the sense that around 28% of the case can be dropped to still maintain 0.7 or higher correlation with their original estimates. The closeness measure (0.5169) for their combined network is already higher than the 0.5 cut-off. The same applies to all edge and strength estimates, with no regard of the network of RRS, SCRS or their combined variant. Betweenness measures are too low to be considered secure to interpret (all below 0.25) meaning that even slight case drops result in losing their stability.

4. Discussion

As (Jones et al., 2018) pointed out, when network modeling is done by using force-directed networks [Fruchterman hivatkozás!!!], node placement is not directly interpretable, as these graphs are rendered to be aesthetic.

Appendix

This combined accuracy estimates for RRS and SCRS include pairs of items from both questionnaires. The last pair of items where the confidence interval does not include 0 edge-weight is the pair of *handle* from RRS and *mistakes* from SCRS. The majority of the more accurate pair estimates are homogeneous in the sense that they are not RRS-SCRS mixed pairs (they are both either from RRS or SCRS). *Handle* belonged to the community of the SCRS nodes in our community analysis.

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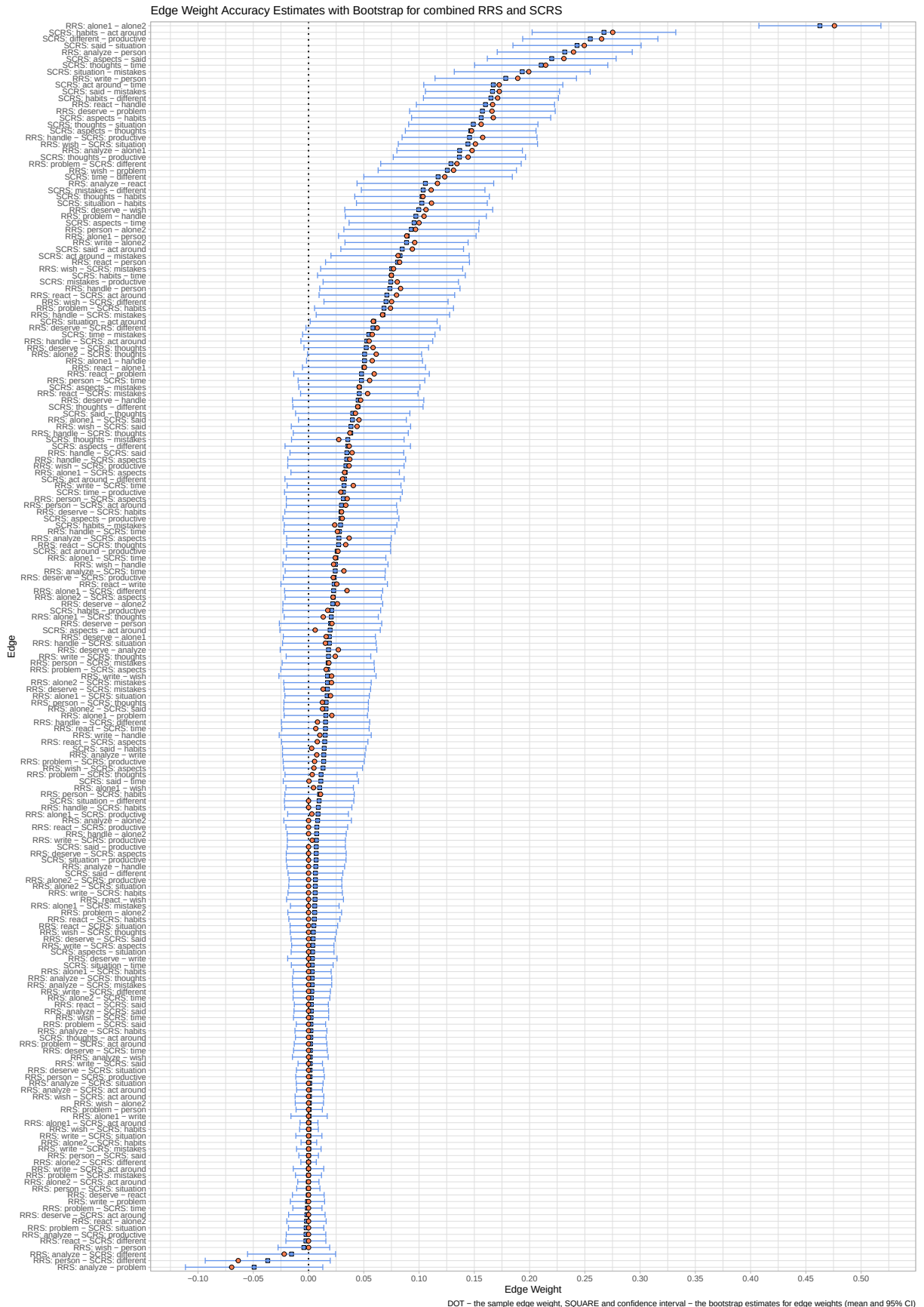


Figure 13: Network accuracy estimate for combined RRS and SCRS